CUSTOMER FAST-TRACK GUIDE

PLANWISE ENTERPRISE

Zero-Friction Quick Start: From Land Intake to Construction Release in 6 Steps

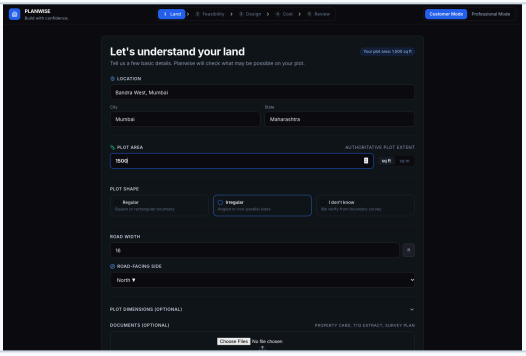
Statutory Notice: Planwise outputs support preliminary planning, feasibility analysis, and computational design synthesis. They do NOT constitute municipal sanctions or licensed structural certifications.

Planwise Enterprise automates residential real estate feasibility, floor plan generation, quantity takeoffs, and contractor budgeting through an integrated 6-step workflow:

Step 1: Enter Land & Property Details

STEP 1: LAND

Input your site address (e.g. Bandra West, Mumbai), select your municipal jurisdiction (Mumbai DCPR 2034 or BBMP 2026), and enter your registered plot area (e.g. 1,500 sq ft). Specify access road widths (existing vs proposed master plan road) to calculate road widening surrender lines.

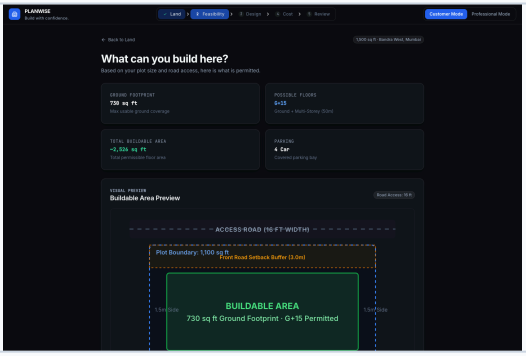


*Action: Click **"Confirm Land & Run Feasibility"** to proceed.*

Step 2: Instant Statutory Feasibility Analysis

STEP 2: FEASIBILITY

The platform applies municipal building bye-laws to compute road widening surrender strips, front/rear/side setbacks, net buildable footprint, permissible Built-Up Area (BUA), and statutory height limits.



*Action: Verify your buildable envelope and click **"Proceed to Step 3: DESIGN"**.*

Fast-Track Architectural Design Synthesis

Step 3A: Specify Living Requirements

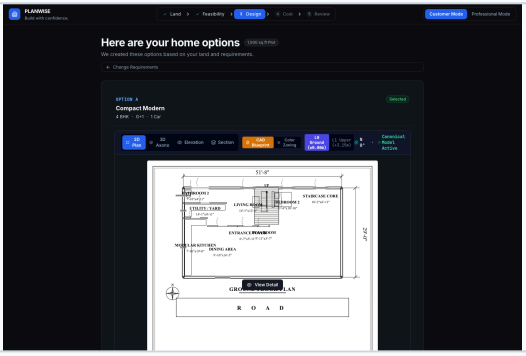
DESIGN BRIEF

Configure your family's room brief: select bedrooms (1 to 5+ BHK), bathrooms (1 to 4+), floor configuration (Single Level vs G+1 Duplex), and architectural style (Contemporary, Minimalist, Traditional, or Modern Indian). Click **"Synthesize Architectural Options"** to run the Google OR-Tools CP-SAT solver.

Step 3B: Compare Pareto-Optimal Layout Candidates

DESIGN OPTIONS

The CP-SAT solver generates collision-free floor plans adhering to NBC 2016 room dimensions, perimeter daylighting rules, and plumbing clustering. Compare candidates across Gross BUA, usable carpet area, spatial efficiency, and estimated budget.

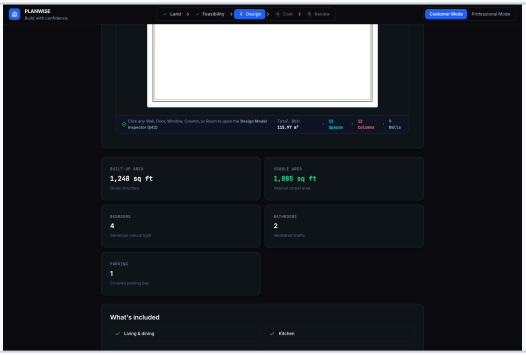


*Action: Select your preferred layout card and click **"Choose This Design & Review Cost"**.*

Step 3C: Read the Architectural Floor Plan Blueprint

CAD BLUEPRINT

Inspect the millimeter-accurate vector CAD drawing: 230mm external masonry, 115mm internal partitions, doors with 90° swing arcs, sliding windows, dimension strings, and room labels. Click any room to adjust dimensions in real time.



Budget Transparency & Build Confirmation

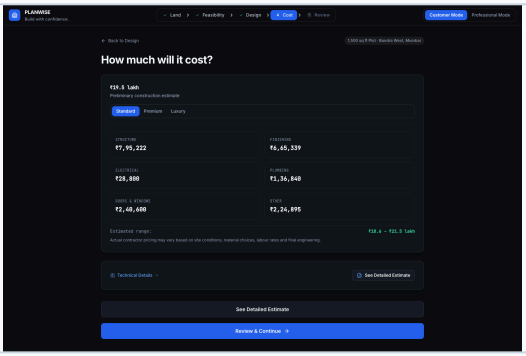
Step 4: Review Itemized Quantities & Turnkey Cost

STEP 4: COST

Quantities are calculated adhering strictly to **IS 1200 measurement standards** (masonry and concrete deducting openings > 0.1 m²; plaster deducting openings > 0.5 m²). The cost calculation executes a transparent financial waterfall:

```
Direct Hard Cost (Sum of BOQ Lines)
  + 8.0% Contractor Prelims & Site Overheads
  + 5.0% Physical Contingency Buffer
= Construction Subtotal
  + 18.0% Statutory Goods & Services Tax (GST)
= TOTAL TURNKEY ESTIMATED BUDGET
```

Toggle between **ECONOMY, STANDARD, PREMIUM,** and **LUXURY** quality tiers to instantly compare finishes.

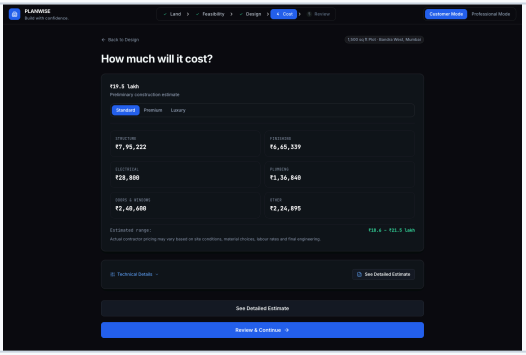


*Action: Confirm quality tier and click **"Proceed to Step 5: BUILD"**.*

Step 5: Lock Build Release & Freeze Design

STEP 5: BUILD

Acknowledge statutory notices regarding geotechnical borehole soil tests and municipal sanction requirements. Click **"Lock Build Release"** to freeze the design with an immutable SHA-256 cryptographic fingerprint.



Professional Review & Document Exports

Step 6: Professional Engineering Review Gates (G0–G6)

STEP 6: ENGINEER

Licensed architects and civil engineers review the design against seven sequential verification gates before construction release:

- **G0 (Site Truth):** Validates boundary coordinates against land survey evidence.
- **G1 (Structural):** Validates Manifold3D watertight 2-manifold solid geometry.
- **G2 (Architectural):** Verifies NBC 2016 room dimensions, ceiling heights, and door clearances.
- **G3 (MEP):** Audits electrical point quotas and wet stack vertical alignment.
- **G4 (Statutory):** Confirms setback clearances, road widening surrender, and FSI caps.
- **G5 (Cost):** Audits itemized BOQ lines against regional CPWD rate catalogs.
- **G6 (Construction Lock):** Final authorization seal gating production drawing release.

Available Project Document Exports

Export Artifact	Format	Description & Use Case
Architectural Drawing Sheets	PDF (A3 / A4)	Site plans, ground floor plans, terrace plans, sections, and door/window schedules.
Customer Home Report	PDF (15 pages)	Executive summary report for homeowners and financing institutions.
Open BIM Model	IFC4 (.ifc)	buildingSMART standard BIM model for Revit, ArchiCAD, and Solibri.
Itemized Bill of Quantities	CSV / JSON	Structured trade quantities adhering to IS 1200 for contractor tendering.
Handoff Release Archive	ZIP Package	Cryptographically sealed bundle containing CBM JSON, IFC4, SVG, BOQ, and validation logs.

System Support: For assistance, run `uvicorn apps.api_gateway.main:app --port 5001` to verify backend health. Refer to the full *Planwise Enterprise User Guide* (PW-DOC-MANUAL-2026) for comprehensive engineering specifications.